

1. Type of IP Block

The proposed IP block is a two-way voltage level shifter. It passes logic signals between the 1.2 V core-logic domain and the 3.3 V I/O domain of an SG13CMOS5L chip. It is a small, self-contained block that uses only the standard thin-oxide (1.2 V) and thick-oxide (3.3 V) MOS transistors in the process. It does not use any SiGe HBTs, any bandgap reference, or any of the shared bias or current-source resources listed in challenge rule 6.

Configuration: the block has four separate channels. Each channel can be set on its own to shift up (1.2 V to 3.3 V) or down (3.3 V to 1.2 V) using its own direction pin. A shared active-low output-enable pin puts the outputs into a high-impedance (tri-state) mode, and a master enable pin turns the block off to save power. A built-in test mode connects an on-chip loop, or ring, path so the block can be checked at full speed across process corners.

Level shifters are one of the most reused blocks in any multi-voltage SoC. Every place where a low-voltage digital core meets 3.3 V pads, sensors, memories, or off-chip parts needs one. So the block meets challenge rule 6 (useful inside an SoC) while taking up very little space on the shared test chip. Only five bond pads are strictly needed, and it uses none of the shared analog resources.

2. I/O List

Signal	Domain / Type	Direction	Function	Pad requirement
VDDH	Power – 3.3 V	—	High-side (3.3 V) supply	Dedicated pad; low series R, Kelvin sense preferred
VDDL	Power – 1.2 V	—	Low-side (1.2 V core) supply	Dedicated pad; low series R, Kelvin sense preferred
VSS	Power – 0 V	—	Common ground reference	Dedicated pad; low series R
A0	1.2 V digital	Bidir	Characterization channel, low-side data	Dedicated low-R pad ($< 50 \Omega$), NOT via shared analog mux
B0	3.3 V digital	Bidir	Characterization channel, high-side data	Dedicated low-R pad ($< 50 \Omega$), NOT via shared analog mux
A1–A3	1.2 V digital	Bidir	Functional channels, low-side data	Via shared digital control/test signals
B1–B3	3.3 V digital	Bidir	Functional channels, high-side data	Via shared digital control/test signals

Signal	Domain / Type	Direction	Function	Pad requirement
DIR<3:0>	1.2 V digital	In	Per-channel direction (0 = A→B up-shift, 1 = B→A down-shift)	Shared control signals
OE_n	1.2 V digital	In	Active-low output enable; tri-states all driven outputs	Shared control signal
EN	1.2 V digital	In	Master enable; low = shutdown / leakage state	Shared control signal
TM	1.2 V digital	In	Test mode; enables on-chip loopback/ring self-test	Shared control signal

Special requirements (challenge rules 17 and 18)

- The two test pins A0 and B0 need a low-resistance path to the pad. These pins carry the high-speed signals used to measure timing. Their path to the pad should have low resistance, below about 50 ohms, and should not run through the shared analog multiplexer. If the resistance is too high, it combines with the pad and probe capacitance and slows the signal edges, which would spoil the delay and rise-and-fall measurements. So we ask for a near-direct, low-resistance connection to these two pads. All the other pins can use the normal shared paths.

3. Functional Description

Each channel holds two one-way translators that share an A-side pin and a B-side pin. The channel's DIR bit picks which one is active and enables its output driver; the other driver stays in high-impedance, so only one side ever drives. This direction-controlled design (instead of auto-sensing) is simple, avoids bus contention, and is easy to check with open-source tools.

Up-conversion path (1.2 V to 3.3 V)

The low-to-high path uses a cross-coupled (latch-type) level shifter made from thick-oxide 3.3 V devices. The 1.2 V input and its complement drive a PMOS latch powered from VDDH, which turns the small 1.2 V swing into a full 0 to 3.3 V output. A thick-oxide driver then buffers it. A skewed pull-down keeps the rising and falling edges balanced to limit duty-cycle distortion at speed.

Down-conversion path (3.3 V to 1.2 V)

The high-to-low path is simpler. A thick-oxide input stage tied to VDDL conditions the 3.3 V input, and a thin-oxide 1.2 V buffer chain restores full core-logic levels and drive strength. The thick-oxide devices at the interface make sure no thin-oxide gate ever sees more than its rated 1.2 V.

Direction, enable, and protection

- Break-before-make switching. When DIR changes, the old driver turns off before the new one turns on, so the two sides never fight or draw shoot-through current.
- Output enable (OE_n). Puts all active drivers into high-impedance for bus sharing, without entering shutdown.
- Master enable (EN). Turns off internal bias and input buffers for a low-leakage shutdown; all outputs go high-impedance.
- Device-stress safety. Every node that can see 3.3 V uses thick-oxide devices, and the 1.2 V side is kept isolated so thin-oxide limits are never exceeded.
- Built-in self-test (TM). Test mode links an up-shifter output to a down-shifter input (and vice versa) to form a loop or ring, giving a one-pin, full-speed pass/fail check across corners without an external pattern generator.

4. Target Specification

Absolute Maximum Ratings

Parameter	Min	Max	Unit
High-side supply, VDDH to VSS	−0.3	+4.0	V
Low-side supply, VDDL to VSS	−0.3	+1.65	V
Voltage on any B-side (3.3 V) pin	−0.3	VDDH + 0.3	V
Voltage on any A-side (1.2 V) pin	−0.3	VDDL + 0.3	V
DC current per data pin	−12	+12	mA
Junction temperature, T _J	—	+125	°C
Storage temperature	−55	+150	°C

Recommended Operating Conditions

Parameter	Min	Typ	Max	Unit
High-side supply, VDDH	2.97	3.30	3.63	V
Low-side supply, VDDL	1.08	1.20	1.32	V
Ambient temperature, T _A (commercial)	0	27	110	°C
Load capacitance per output, C _L	—	5	15	pF
Input edge rate at A/B data pin	—	—	5	ns

DC Electrical Characteristics

Parameter	Symbol	Min	Typ	Max	Unit
High-level input voltage (A-side)	V _{IH_L}	0.65 · VDD _L	—	—	V
Low-level input voltage (A-side)	V _{IL_L}	—	—	0.35 · VDD _L	V
High-level input voltage (B-side)	V _{IH_H}	0.65 · VDD _H	—	—	V
Low-level input voltage (B-side)	V _{IL_H}	—	—	0.35 · VDD _H	V
High-level output voltage	V _{OH}	VDD − 0.1	—	—	V
Low-level output voltage	V _{OL}	—	—	0.1	V
Output drive current (at V _{OH} /V _{OL})	I _{OH} /I _{OL}	±2	—	—	mA
Input leakage current	I _{IN}	—	—	±1	μA
Tri-state output leakage	I _{OZ}	—	—	±1	μA
Quiescent current, enabled (per rail)	I _Q	—	1	5	μA
Shutdown current, EN = 0 (per rail)	I _{SD}	—	0.1	1	μA

AC / Timing Characteristics

Per channel, C_L = 5 pF, over the recommended operating range unless noted.

Parameter	Symbol	Min	Typ	Max	Unit
Propagation delay, up-shift (1.2→3.3)	tPD_UP	—	1.5	3.0	ns
Propagation delay, down-shift (3.3→1.2)	tPD_DN	—	1.0	2.5	ns
Low-high / high-low delay skew	tSKLH	—	0.2	0.6	ns
Channel-to-channel skew	tSKCH	—	0.2	0.5	ns
Output rise / fall time (20–80%)	tR/tF	—	1.0	2.0	ns
Max data rate (guaranteed over PVT)	fMAX	100	200	—	Mbps
Max data rate (typical, stretch goal)	fMAX,typ	—	500	—	Mbps
Duty-cycle distortion at fMAX	DCD	—	±0.2	±0.5	ns
Direction turnaround time (DIR to valid)	tDIR	—	4	10	ns
Output-enable / disable time	tOE/tOD	—	2	6	ns
Enable-to-ready (wake from shutdown)	tWAKE	—	0.5	2	μs

4.5 Dynamic Power

Most of the dynamic current comes from charging the 3.3 V output node, which follows the usual CV^2f rule. The goal is no more than 8 pJ per bit transition (up-shift, $CL = 5$ pF). At 100 MHz, that works out to about 0.8 mA per channel from VDDH, and it rises in step with data rate and load. Current draw should be smooth and predictable, with no steady short-circuit path except during the brief break-before-make gap.

5. Test Plan (Measurement-Based Validation)

5.1 DC and functional

1. **Logic sweep:** with the block on, step each channel through all four DIR/data combos both directions; confirm correct translation and VOH/VOL/VIH/VIL (section 4.3).
2. **Tri-state and enable:** assert OE_n, confirm high-Z (IOZ); toggle EN, confirm shutdown current (ISD) on each rail.
3. **Leakage:** measure IQ and ISD per rail via Kelvin connections, across temperature.

5.2 Timing and high-speed

4. **Delay and edges:** drive a ≤ 200 ps edge into A0/B0; capture the opposite pad on a ≥ 4 GHz scope; extract tPD_UP, tPD_DN, tR, tF, and skew.
5. **Max data rate / DCD:** sweep rate with a PRBS pattern; record eye diagrams; fMAX is where the eye mask or DCD limit first fails, both directions.
6. **Direction turnaround:** toggle DIR while driving both sides; measure tDIR and confirm break-before-make (no VDDH contention spike).
7. **Output-enable timing:** measure tOE/tOD from OE_n into a fixed pull load.

5.3 PVT corners and reliability

8. **Supply corners:** repeat 5.1 and key 5.2 tests at VDDL/VDDH min/typ/max.
9. **Temperature:** repeat core tests over 0–110 C, and –40 to +125 C for characterization (rule 16).
10. **Built-in self-test:** run the on-chip loopback/ring at each corner as a pass/fail gate and delay cross-check.
11. **Sim correlation:** overlay measured delay, fMAX, and current on post-layout ngspice; one summary plot (rule 13).

5.4 Pass/fail

Passes if, across the commercial range, it meets every section 4 limit, shows no bus contention on direction changes, and stays below ISD. Scripts, board/probe setup, and extraction flow go in the public repo for open-source reproduction (rules 9–15).